

Balanced Sigma-MGFs for Tensor-Product Local Unitary Groups

Proof, exact formulas, and direct matrix verification

Reproducibility note

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Outcome. For $G = \prod_{a=1}^k U(d_a)$ on $V = \bigotimes_{a=1}^k \mathbb{C}^{d_a}$, the ordinary one-sided sigma-MGF is exactly 1. The nontrivial object is the balanced two-sided series involving both V and V^* . Its stable power-sum coefficient is z_λ^{k-2} , its bipartite single-state coefficient is a restricted partition number, and its multilinear coefficient is an exact product of Schur–Weyl centralizer dimensions. A reproducible marimo notebook constructs the Kronecker-product matrices and compares direct Haar estimates with these formulas. All 46 numerical comparisons, 4 construction checks, and 5 scalar-center witnesses pass.

1 Representation and the correction at the outset

Put $V_a = \mathbb{C}^{d_a}$ and

$$G = U(d_1) \times \cdots \times U(d_k), \quad V = V_1 \otimes \cdots \otimes V_k.$$

The representation is the concrete matrix homomorphism

$$\rho(g_1, \dots, g_k) = g_1 \otimes \cdots \otimes g_k \in U(D), \quad D = \prod_{a=1}^k d_a. \quad (1)$$

For a partition $\tau = (\tau_1, \dots, \tau_\ell)$, write

$$\mathrm{Sym}^\tau V = \bigotimes_{j=1}^\ell \mathrm{Sym}^{\tau_j} V, \quad |\tau| = \sum_j \tau_j.$$

The ordinary coefficient is

$$[m_\tau] \mathrm{SMG}_{G,V} = \dim(\mathrm{Sym}^\tau V)^G.$$

Proposition 1 (one-sided collapse).

$$\boxed{\mathrm{SMG}_{G,V} = 1.} \quad (2)$$

Proof. For $z \in U(1)$ the group contains $(zI_{d_1}, I_{d_2}, \dots, I_{d_k})$, whose matrix in (1) is zI_V . It acts on $\mathrm{Sym}^\tau V$ as $z^{|\tau|}I$. If $|\tau| > 0$, one can choose z with $z^{|\tau|} \neq 1$, so a fixed vector must be zero. Only degree zero remains. $\square$

The same center explains why local-unitary entanglement invariants use a state ψ and its conjugate $\bar{\psi}$: the weights z and z^{-1} cancel only when the two total degrees agree.

2 The balanced series and Molien formula

Introduce independent alphabets $\mathbf{s}$ and $\mathbf{t}$ and define

$$\mathcal{B}_{G,V}(\mathbf{s}, \mathbf{t}) = \int_G \text{Exp}_\sigma(X_V(g)h_1(\mathbf{s})) \text{Exp}_\sigma(X_{V^*}(g)h_1(\mathbf{t})) dg. \quad (3)$$

Coefficient extraction gives

$$\mathcal{B}_{G,V} = \sum_{\alpha, \beta} b_{\alpha, \beta} m_\alpha(\mathbf{s}) m_\beta(\mathbf{t}), \quad (4)$$

$$b_{\alpha, \beta} = \dim \left(\text{Sym}^\alpha V \otimes \text{Sym}^\beta V^* \right)^G = \dim \text{Hom}_G(\text{Sym}^\beta V, \text{Sym}^\alpha V). \quad (5)$$

The center now gives the balanced selection rule

$$\boxed{b_{\alpha, \beta} = 0 \quad \text{unless} \quad |\alpha| = |\beta|.} \quad (6)$$

For finitely many variables, (3) is the balanced Molien integral

$$\mathcal{B}_{G,V} = \int_G \prod_i \det(I - s_i \rho(g))^{-1} \prod_j \det(I - t_j \rho(g)^{-1})^{-1} dg. \quad (7)$$

If $x_1^{(a)}, \dots, x_{d_a}^{(a)}$ are the eigenvalues of g_a , then the eigenvalues of $\rho(g)$ are $x_{i_1}^{(1)} \cdots x_{i_k}^{(k)}$. Consequently (7), together with the Weyl integration formula for each local factor, is an exact constant-term prescription in every dimension.

3 Stable power-sum formula

For every positive integer r , Kronecker products satisfy the directly testable identity

$$\text{Tr}(\rho(g)^r) = \prod_{a=1}^k \text{Tr}(g_a^r). \quad (8)$$

Thus, for a partition λ , $p_\lambda(\rho(g)) = \prod_a p_\lambda(g_a)$, and product Haar measure gives

$$\mathbb{E}_G \left[p_\lambda(\rho(g)) \overline{p_\mu(\rho(g))} \right] = \prod_{a=1}^k \mathbb{E}_{U(d_a)} \left[p_\lambda(g_a) \overline{p_\mu(g_a)} \right]. \quad (9)$$

Let $z_\lambda = \prod_i i^{m_i(\lambda)} m_i(\lambda)!$. Schur orthogonality, or equivalently the stable Hall inner product, gives

$$\mathbb{E}_{U(d)} [p_\lambda(U) \overline{p_\mu(U)}] = \delta_{\lambda\mu} z_\lambda \quad \text{when } d \geq |\lambda|, |\mu|. \quad (10)$$

Combining (9) and (10), if $N \leq \min_a d_a$ then

$$\mathbb{E}_G [p_\lambda(\rho) \overline{p_\mu(\rho)}] = \delta_{\lambda\mu} z_\lambda^k \quad (|\lambda|, |\mu| \leq N). \quad (11)$$

Each determinant exponential contributes a factor p_λ/z_λ . Therefore through bidegree (N, N) ,

$$\boxed{\mathcal{B}_{G,V} \equiv \sum_{|\lambda| \leq N} z_\lambda^{k-2} p_\lambda(\mathbf{s}) p_\lambda(\mathbf{t}).} \quad (12)$$

For $k = 1, 2, 3$, the coefficients are respectively $z_\lambda^{-1}, 1, z_\lambda$.

4 Exact single-state coefficients

For partitions $\lambda^{(1)}, \dots, \lambda^{(k)} \vdash m$, define the generalized Kronecker coefficient

$$g(\lambda^{(1)}, \dots, \lambda^{(k)}) = \dim \left([\lambda^{(1)}] \otimes \dots \otimes [\lambda^{(k)}] \right)^{S_m}.$$

Applying Schur–Weyl duality to each $V_a^{\otimes m}$ and then taking diagonal S_m -invariants proves

$$\mathrm{Sym}^m(V_1 \otimes \dots \otimes V_k) \cong \bigoplus_{\lambda^{(1)}, \dots, \lambda^{(k)} \vdash m} g(\lambda^{(1)}, \dots, \lambda^{(k)}) \bigotimes_{a=1}^k S_{\lambda^{(a)}} V_a, \quad (13)$$

where terms with $\ell(\lambda^{(a)}) > d_a$ vanish. Schur orthogonality now gives the all-dimensional identity

$$b_{(m), (m)} = \sum_{\substack{\lambda^{(a)} \vdash m \\ \ell(\lambda^{(a)}) \leq d_a}} g(\lambda^{(1)}, \dots, \lambda^{(k)})^2. \quad (14)$$

For $k = 2$, $g(\lambda, \mu) = \delta_{\lambda\mu}$. Hence

$$b_{(m), (m)} = \#\{\lambda \vdash m : \ell(\lambda) \leq \min(d_1, d_2)\}. \quad (15)$$

This is $p(m)$ when both dimensions are at least m . In stable range, setting each alphabet in (12) to one variable yields the equivalent formula

$$b_{(m), (m)} = \sum_{\lambda \vdash m} z_\lambda^{k-2}. \quad (16)$$

For completeness, general $b_{\alpha, \beta}$ also has an exact finite formula. Decompose each $\mathrm{Sym}^{\alpha_j} V$ by (13), then combine local Schur functors with Littlewood–Richardson coefficients. If the resulting multiplicity of $\bigotimes_a S_{\mu^{(a)}} V_a$ is $M_\alpha(\mu^{(1)}, \dots, \mu^{(k)})$, then

$$b_{\alpha, \beta} = \sum_{\substack{\mu^{(1)}, \dots, \mu^{(k)} \\ \ell(\mu^{(a)}) \leq d_a}} M_\alpha(\mu^{(1)}, \dots, \mu^{(k)}) M_\beta(\mu^{(1)}, \dots, \mu^{(k)}). \quad (17)$$

This separates the two combinatorial ingredients: generalized Kronecker coefficients enter the symmetric powers of a tensor product, while Littlewood–Richardson coefficients enter products of symmetric powers.

5 Exact multilinear coefficients

For $\alpha = \beta = (1^r)$, $\mathrm{Sym}^{(1^r)} V = V^{\otimes r}$, so

$$b_{(1^r), (1^r)} = \dim \mathrm{End}_G(V^{\otimes r}). \quad (18)$$

Because the local factors act independently,

$$\mathrm{End}_G(V^{\otimes r}) \cong \bigotimes_{a=1}^k \mathrm{End}_{U(d_a)}(V_a^{\otimes r}).$$

If f^λ is the number of standard Young tableaux of shape λ , Schur–Weyl duality and the hook-length formula give

$$b_{(1^r), (1^r)} = \prod_{a=1}^k \sum_{\substack{\lambda \vdash r \\ \ell(\lambda) \leq d_a}} (f^\lambda)^2. \quad (19)$$

When every $d_a \geq r$, the inner sum is $|S_r| = r!$, and therefore

$$b_{(1^r), (1^r)} = (r!)^k. \quad (20)$$

This is the dimension of the stable family of contraction diagrams indexed by k independent permutations in S_r .

6 Verification design

The accompanying implementation uses two independent routes.

Exact route. Integer partitions are generated recursively. The hook-length formula computes f^λ , which is substituted into (19). Equation (15) is evaluated by restricted partition counting, and (16) by direct computation of z_λ . No sampled matrices enter these tables.

Direct matrix route. For every local factor, a complex Gaussian matrix Z is QR-factorized and the diagonal phases of R are removed. The resulting Q is Haar distributed on $U(d_a)$. The code explicitly forms $\rho = Q_1 \otimes \cdots \otimes Q_k$, checks $\rho^* \rho = I$, and evaluates its power traces. Newton’s recurrence

$$mh_m(\rho) = \sum_{j=1}^m \text{Tr}(\rho^j) h_{m-j}(\rho) \quad (21)$$

then produces characters of $\text{Sym}^m V$ directly. The sampled target quantities are

$$\begin{aligned} \mathbb{E} \prod_j h_{\tau_j}(\rho) & \quad (\text{ordinary one-sided}), \\ \mathbb{E}[p_\lambda(\rho) \overline{p_\mu(\rho)}] & \quad (\text{balanced power-sum}), \\ \mathbb{E}|h_m(\rho)|^2 & \quad (\text{single-state}), \\ \mathbb{E}|\text{Tr} \rho|^{2r} & \quad (\text{multilinear}). \end{aligned}$$

The last two are character inner products, hence exactly the invariant multiplicities in (14) and (19). Numerical integration is corroboration, not a replacement for the proofs above.

7 Results

Each case used 6,000 deterministic-seed Haar samples. A numerical row passes when the absolute error is at most six empirical standard errors plus a small fixed rounding margin. Matrix unitarity and trace factorization were also checked at tolerance 10^{-11} .

Representation	Target	Direct estimate	Formula	Std. error	Result
$U(3)$	$ h_3 ^2$	1.0367	1	0.0253	PASS
$U(3)$	$ \text{Tr } \rho ^4$	2.0318	2	0.0593	PASS
$U(2) \times U(3)$	$ h_2 ^2$	1.9533	2	0.0546	PASS
$U(2) \times U(3)$	$ p_{(2)} ^2$	3.8911	4	0.0638	PASS
$U(2) \times U(2)$	$ h_3 ^2$	2.0797	2	0.1219	PASS
$U(2)^3$	$ h_2 ^2$	4.0815	4	0.1641	PASS
$U(2)^3$	$ \text{Tr } \rho ^4$	8.2364	8	0.6379	PASS
$U(2)^3$	$p_{(2)}\overline{p_{(1,1)}}$	0.0032	0	0.0284	PASS

The exact calculations make the stable-range boundary visible:

Dimensions	Sector	Degree	Exact	Stable expression
(2, 2)	single-state	3	2	$p(3) = 3$ (not applicable)
(2, 2)	multilinear	3	25	$(3!)^2 = 36$ (not applicable)
(2, 3)	multilinear	3	30	$(3!)^2 = 36$ (not applicable)
(2, 2, 2)	multilinear	3	125	$(3!)^3 = 216$ (not applicable)

Thus the dimension restrictions are not cosmetic: applying a stable formula outside $r, m \leq \min_a d_a$ produces wrong answers.

The scalar-center test separately constructs, for each degree $1 \leq n \leq 5$, $z = e^{2\pi i/(n+1)}$ and the group matrix $\rho(zI_2, I_3) = zI_6$. Its action on degree n is $z^n \neq 1$ in every case, giving five direct matrix witnesses for Proposition 1.

8 Reproduction and scope

The folder `for_this_guy/local_unitary_tensor_products` contains:

- `notebook.py`, the interactive marimo report;
- `verification.py`, the direct and exact implementation;
- `test_verification.py`, deterministic regression tests;
- this L^AT_EX source and its compiled PDF.

The notebook exposes the sample count and reports every tested coefficient, not only the representative rows printed here. Its global $U(3)$ case is explicitly labeled as a comparator; it is not conflated with the local tensor-product groups. All local-unitary cases remain grouped in this single family because they are specializations of the same representation (1) and the same proof.